

to list all the AWS images we created:

```
$ ops image list -t aws -z us-east-1
```

In the screenshot given in Figure 6, do note the image name under the column 'Name'.

Once we have the image name, we can use the same to create our instance. To do so, we should run the command given below. This will create the EC2 instance for us within a VPC and will associate security groups.

```
$ ops instance create node-image -t aws -z us-east-1
```

It will take a few seconds to launch and boot the instance, and we will have an output somewhat similar to the screenshot in Figure 7.

There is an Ops command that will help us to list all instances running on AWS, and will also provide us information related to our instance, like the instance status and public IPs.

```
$ ops instance list -t aws -z us-east-1
```

Now, to access the application we need to visit the application URL from a browser. For this particular demo or proof-of-concept, the public IP assigned to our instance is our URL and the application is accessible over `http://<public-ip>:3000`. That's it folks! We now have a Nanos based Node.js unikernel instance running our pocket calculator app on AWS.

Port 3000 will not be added automatically to the ingress or inbound rules within the AWS security group. We will have to add it manually.

Although unikernels are an emerging technology and are growing fast, the concept or technology is not new. This revolution was brewing since the mid-90s but did not gain traction because we were dealing with physical hardware and the elasticity of the cloud was alien to us. The evolution of cloud computing paved

```
shambo@unikern:/opt/calc-app$ ops image create -t aws -c config.json --package eyberg/node:v16.5.0
100% [1s:0s] (466/466, 7 it/s)
aws image 'node-image' created...
```

Figure 5: An AWS image successfully created from the Ops Node.js package

```
shambo@unikern:/opt/calc-app$ ops image list -t aws -z us-east-1
```

AMIID	AMINAME	NAME	STATUS	CREATED
ami-075ab1bd182e15498	node-image1655223883207704466	node-image	available	1 minute ago

Figure 6: Output after the execution of Ops image list command

```
shambo@unikern:~$ ops instance create node-image -t aws -z us-east-1
Created security_group sg-027b998bde3c87403 with VPC vpc-05829ba2421bcab6a.
```

Figure 7: Creation of EC2 instance and security groups

```
shambo@unikern:/opt/calc-app$ ops instance list -t aws -z us-east-1
```

NAME	ID	STATUS	CREATED	PRIVATE IPS	PUBLIC IPS
node-image-1656674944	i-00a831e4220622ba8	running	2022-07-01 11:29:11 +0000 UTC	172.31.52.18	54.175.172.97

Figure 8: List of instances along with information related to each instance

the path of success for technologies like containerisation and unikernels. Unikernel's minimalist approach and security by design hold the future where security, cost optimisation and availability will gain more importance. Having said that, we may not want to run our production-ready microservices based applications as unikernels yet; containers are still the preferred solution. But it is quite evident that containers are walking the unikernel path — efforts are

being made to make container images leaner and meaner to reduce the attack surface. Efforts are also being made to minimise the interaction of containers with the host operating system for better isolation. Unikernels may not seem like a standard solution as they are not widely adopted right now. But as a DevOps or a DevSecOps professional, a microservices developer, or a technology enthusiast it will definitely be worth your while to explore them. 

References

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- What is Nanos? <https://Nanos.org/>
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- Introduction to Ops, <https://docs.ops.city/ops/>

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